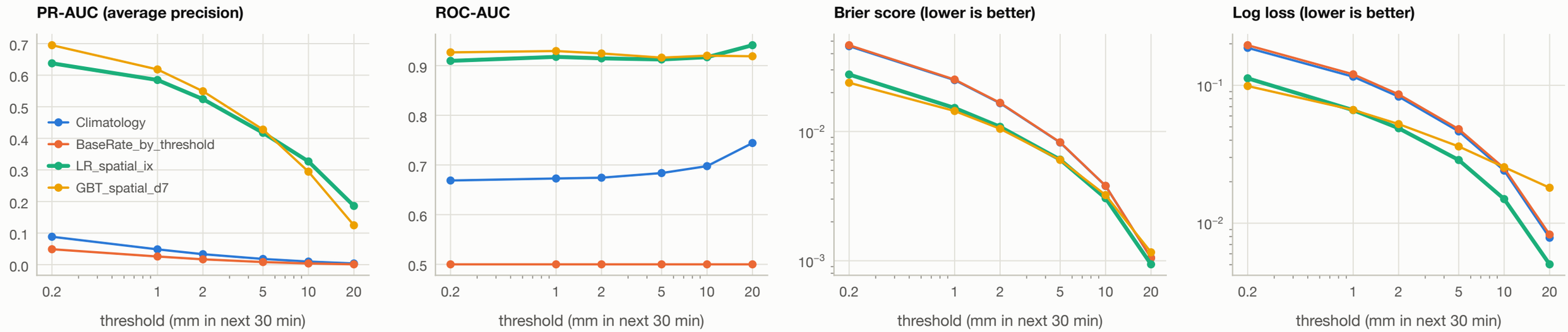
# Skill is highest for the common, light-rain thresholds and falls as the event gets rarer

2024 test set, one model per colour. PR-AUC must be read against prevalence, which drops from ≈5% (0.2 mm) to ≈0.1% (20 mm).

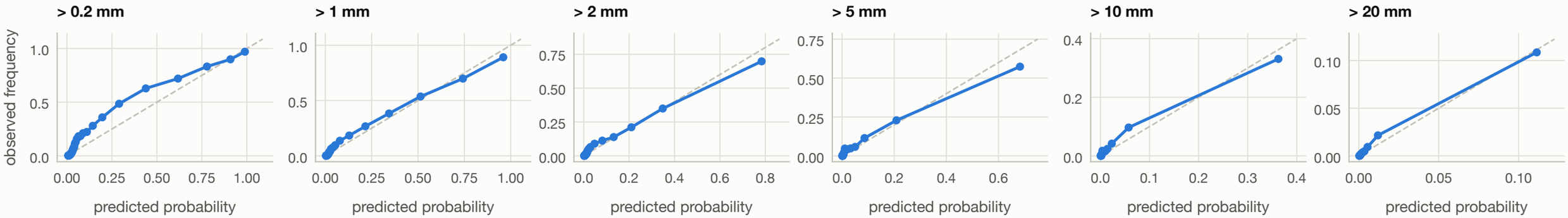



Source: data.gov.sg rainfall, 2017–2024; Spark MLlib. Target: rain accumulated in the 30 min AFTER the prediction time exceeds the threshold (strict >).

Train 2017–2022 · model choice on 2023 · 2024 scored once after selection.

# Predicted probabilities track observed frequencies

Reliability on 2024 for the selected model (LR_spatial_ix); dashed line = perfect calibration. Equal-count-style bins from the evaluation code.

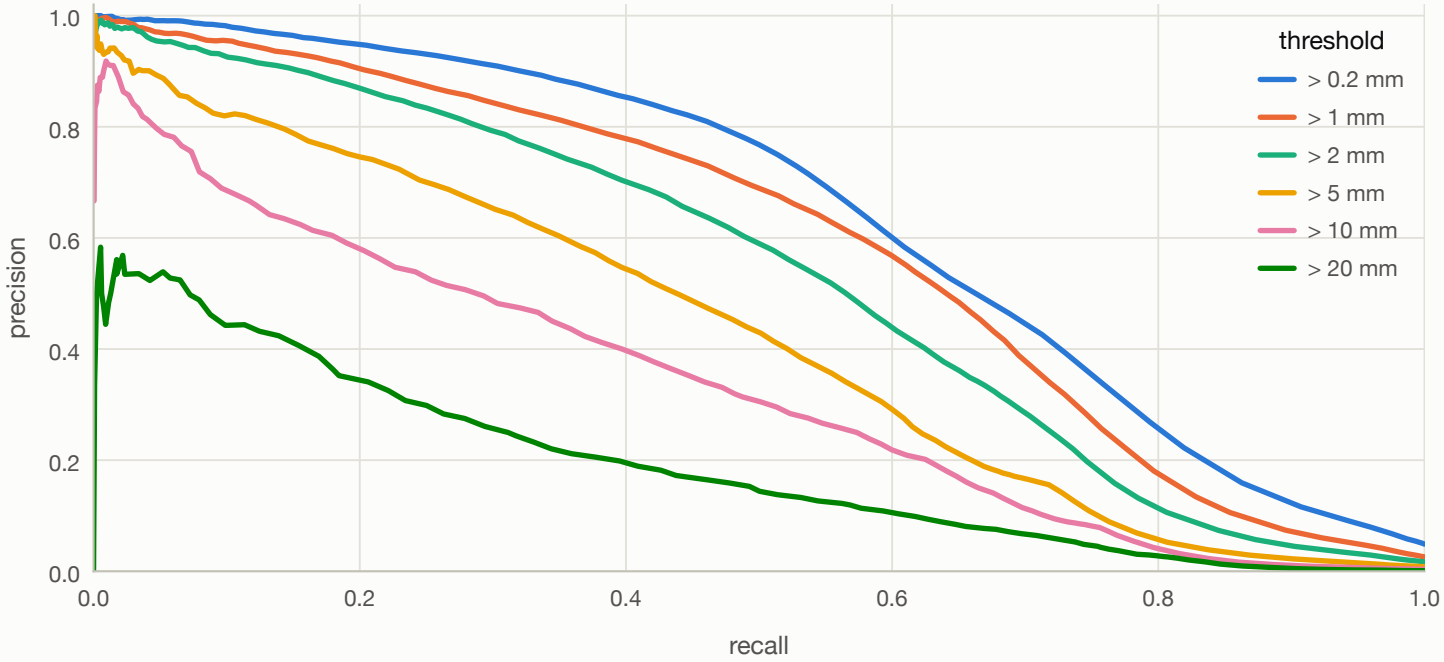



Source: data.gov.sg rainfall, 2017–2024; Spark MLlib. Target: rain accumulated in the 30 min AFTER the prediction time exceeds the threshold (strict >).

Train 2017–2022 · model choice on 2023 · 2024 scored once after selection.

# Precision falls quickly as recall is pushed up, most steeply for heavy rain

Precision–recall curves on 2024, selected model (LR_spatial_ix).

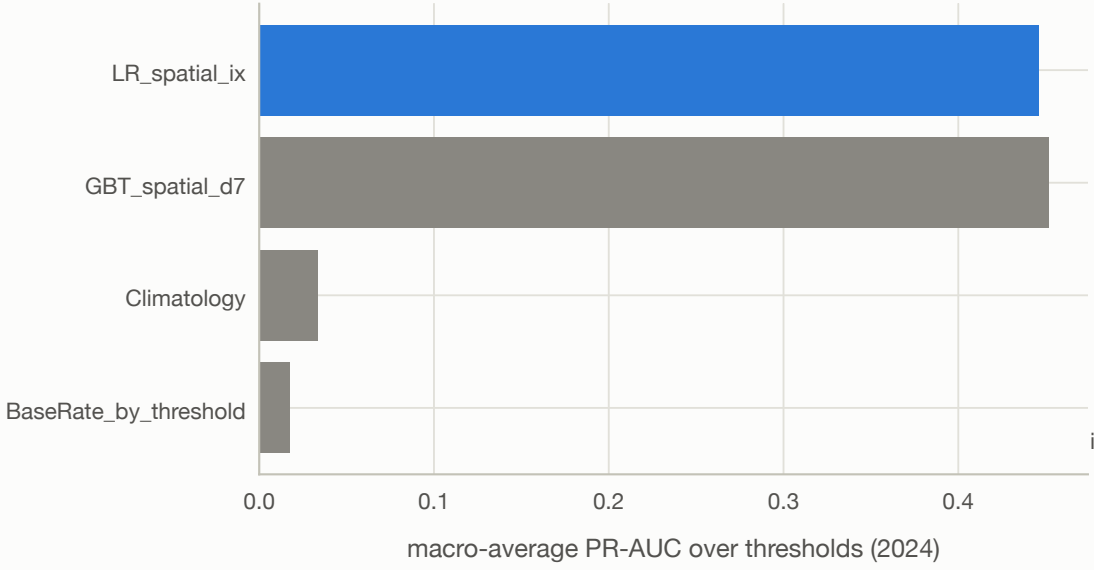


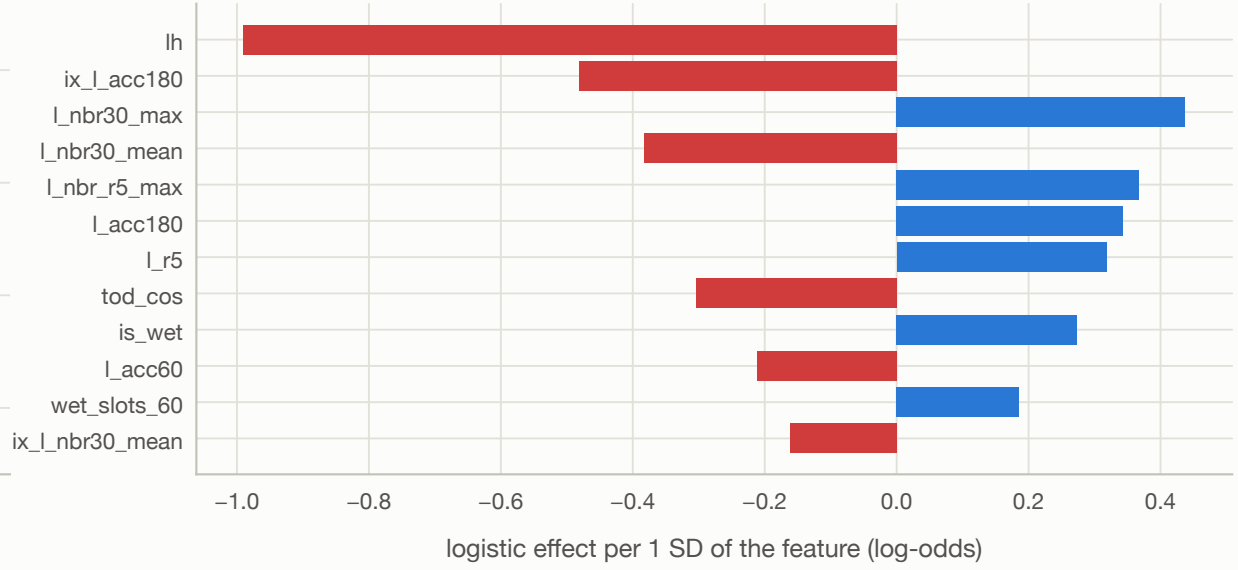

Source: data.gov.sg rainfall, 2017–2024; Spark MLlib. Target: rain accumulated in the 30 min AFTER the prediction time exceeds the threshold (strict >).

Train 2017–2022 · model choice on 2023 · 2024 scored once after selection.

# Recent rain nearby and the threshold itself dominate; model comparison on the left

Left: macro PR-AUC on the untouched 2024 test set. Right: largest standardized effects of the best logistic model (fit on training years).



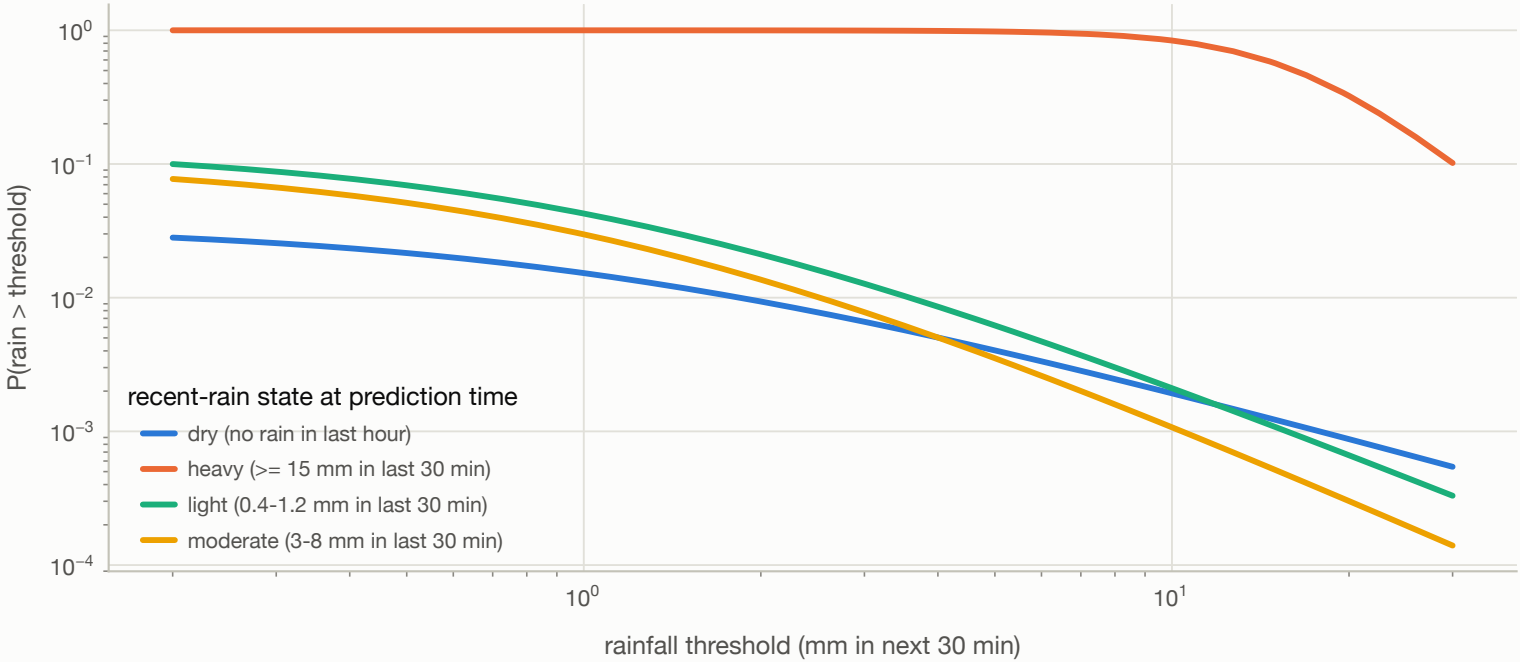




Source: data.gov.sg rainfall, 2017–2024; Spark MLlib. Target: rain accumulated in the 30 min AFTER the prediction time exceeds the threshold (strict >).

Train 2017–2022 · model choice on 2023 · 2024 scored once after selection.

# The model answers for any threshold, and the probability falls as the threshold rises

Station S228, 2024 afternoon prediction times chosen by recent-rain state. Same station, four conditions, 36+ thresholds.




Source: data.gov.sg rainfall, 2017–2024; Spark MLlib. Target: rain accumulated in the 30 min AFTER the prediction time exceeds the threshold (strict >).

Train 2017–2022 · model choice on 2023 · 2024 scored once after selection.